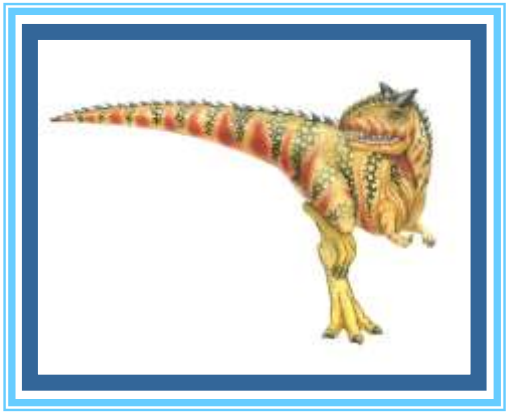
Chapter 12: Mass-Storage Structure



Silberschatz, Galvin and Gagne ©2013

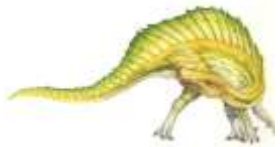


Chapter 12: Mass-Storage Structure

- 12.1 Overview of Mass Storage Structure
- 12.4 Disk Scheduling
- 12.7 RAID Structure

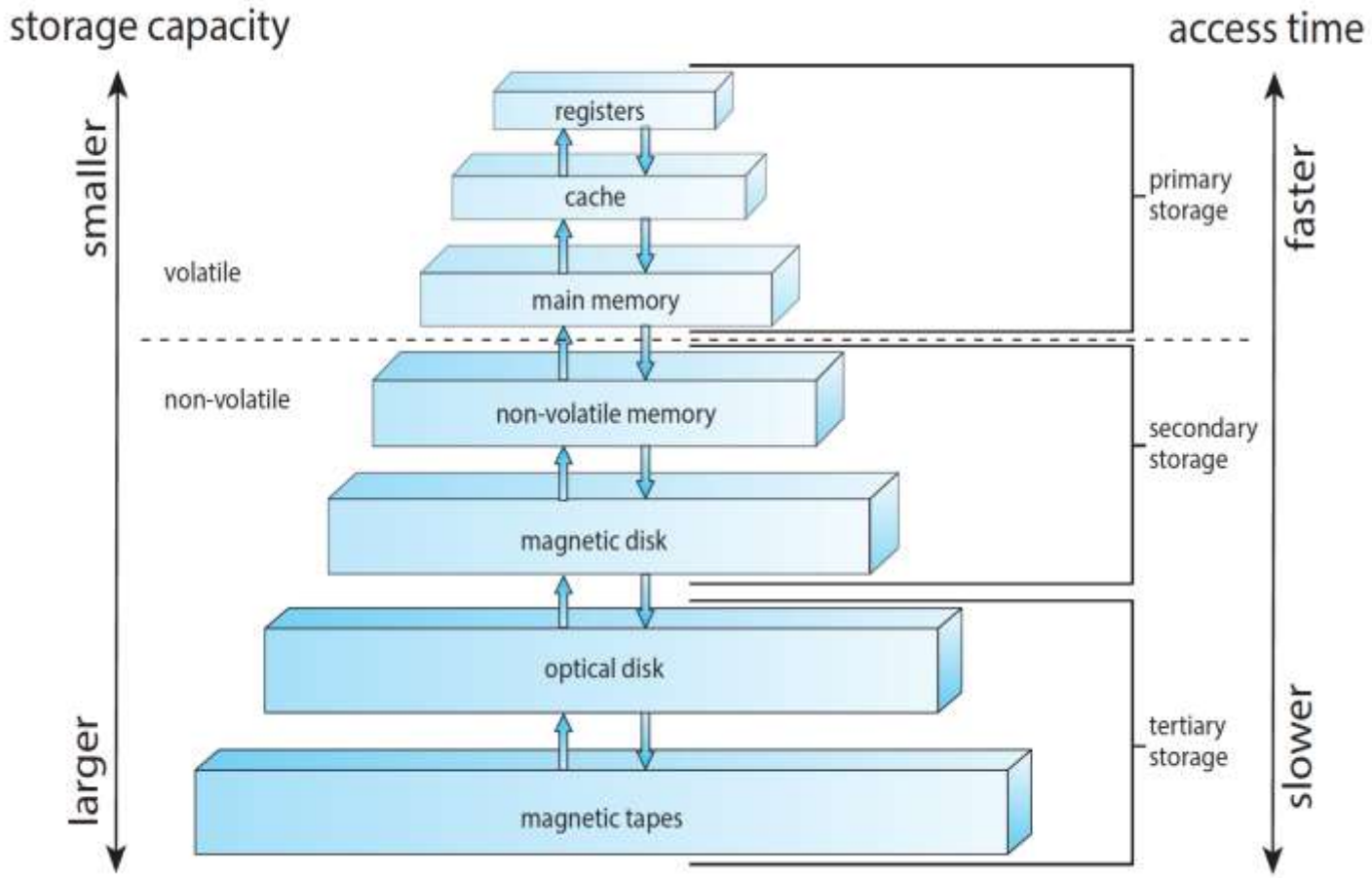
OBJECTIVE

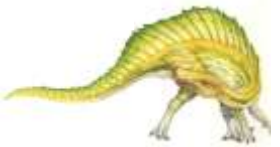
- To describe the physical structure of secondary storage devices and its effects on the uses of the devices
- To explain the performance characteristics of mass-storage devices
- To evaluate I/O scheduling algorithms
- To discuss operating-system services provided for mass storage, including RAID





Storage-Device Hierarchy

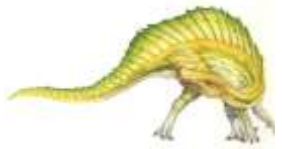






Overview of Mass-Storage Structure

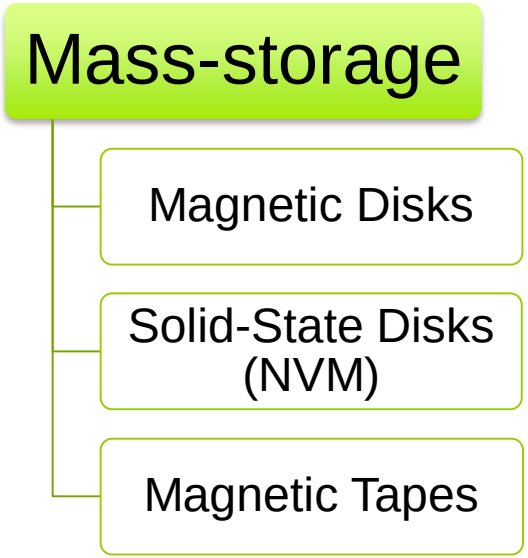
- The bulk of secondary storage for modern computer are
  - Hard disk drives (HDD)
  - Nonvolatile Memory (NVM)
- The most common tertiary storage is magnetic tape.
- In this chapter we describe:
  - The basic mechanism of those devices
  - How operating systems translate their physical properties to logical storage via address mapping.





Mass-Storage Structure

Overview







Mass-Storage Structure

Magnetic disks

- **Magnetic disks** provide bulk of secondary storage of modern computers.
- Disk **platter**: each one has a flat circular shape.
  - Diameter range: 1.8” to 3.5” (inches).
- The two surfaces of a platter are covered with a magnetic materials.

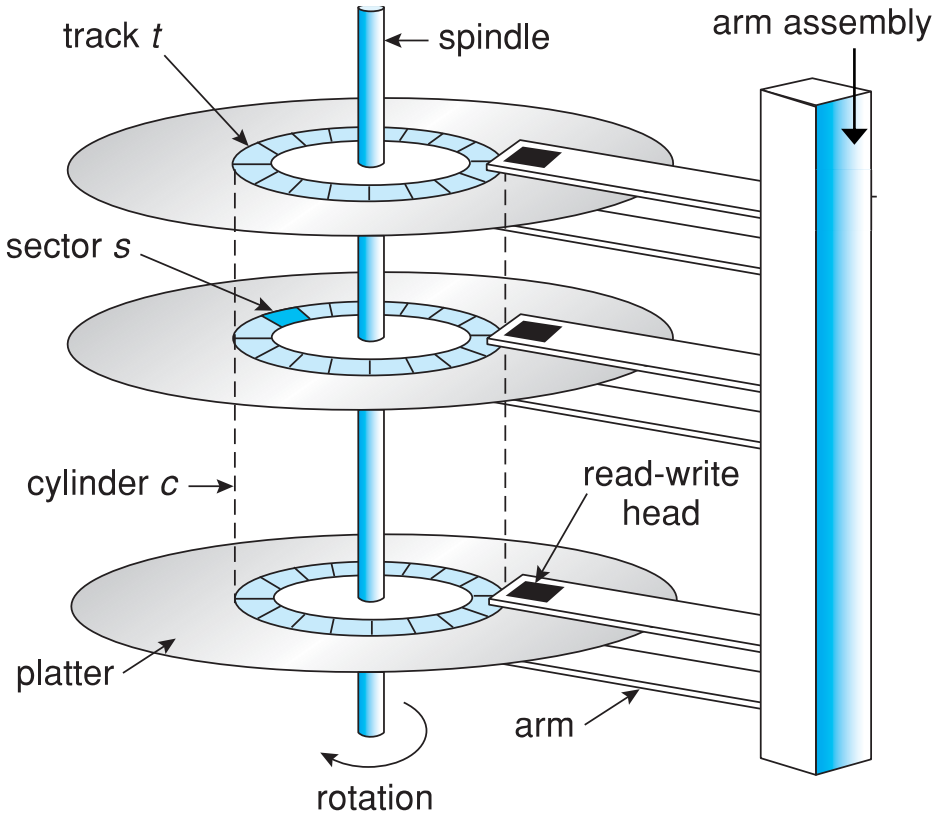


Figure 12.1 Moving-head Disk Mechanism.



Mass-Storage Structure

Magnetic disks

- Storage capacity measured in *gigabytes*: Range from 30*GB* to 3*TB* per drive.
- The read-write **heads** attached to a **disk arm** that moves all the heads as a unit.
- The surface of a platter is logically divided into circular **tracks**, which subdivided into **sectors**.

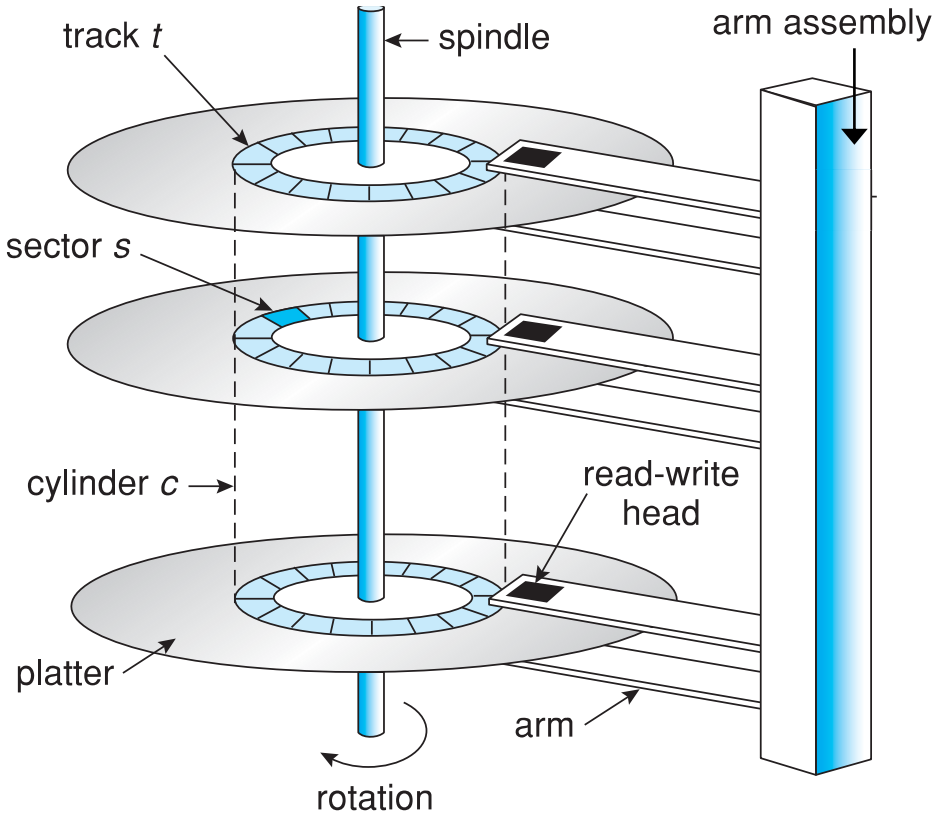


Figure 12.1 Moving-head Disk Mechanism.



Mass-Storage Structure

Magnetic disks

- The sets of tracks at one arm position makes up a **cylinder**.
- A drive motor spins (rotate) it at high speed: 60 to 250 times per second.
- Specified in term of **Rotations Per Minute** (**RPM**) → 5400, 7200, 10000, and 15000 RPM.

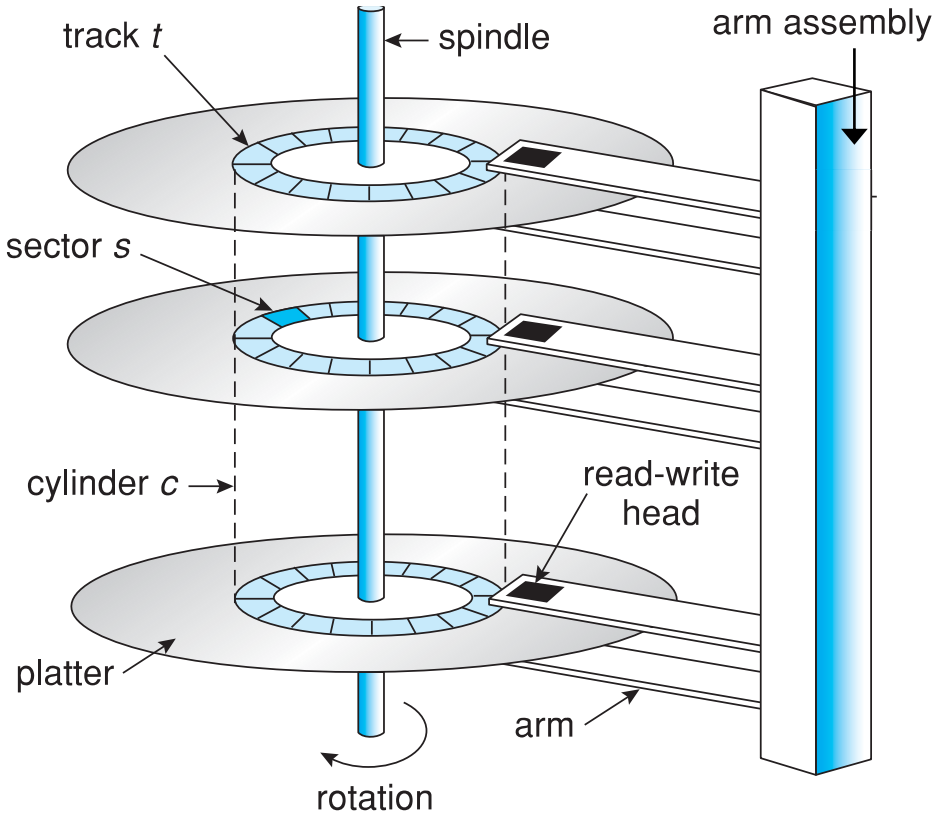


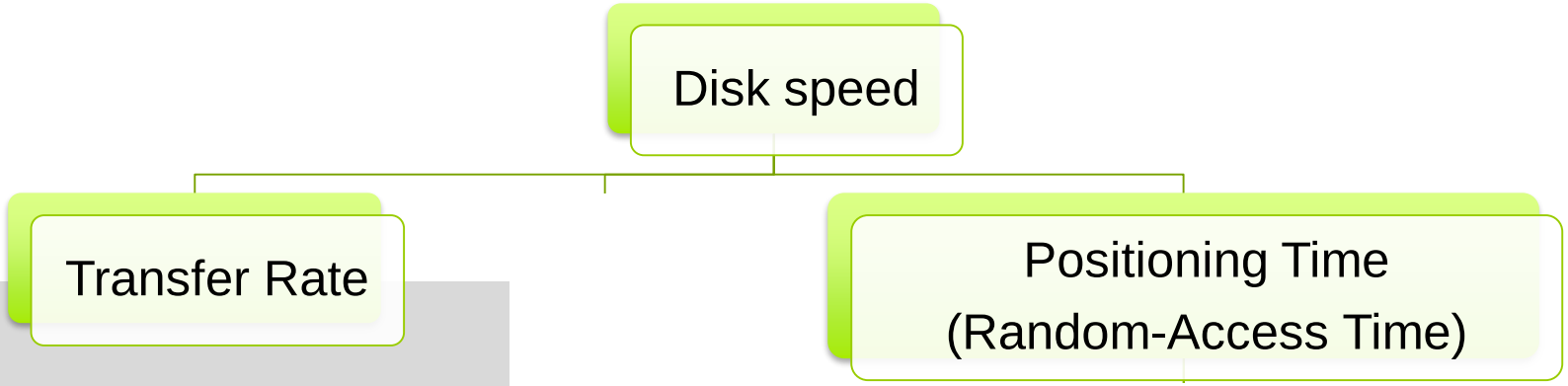
Figure 12.1 Moving-head Disk Mechanism.



Mass-Storage Structure

Magnetic disks

- Two parts of disks speed:



→ rate at which data flow between drive and computer

→ time to move disk arm to desired cylinder.

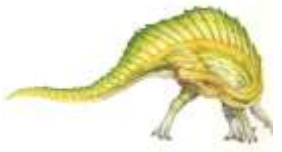
→ time for desired sector to rotate under the disk head.



Mass-Storage Structure

Magnetic disks

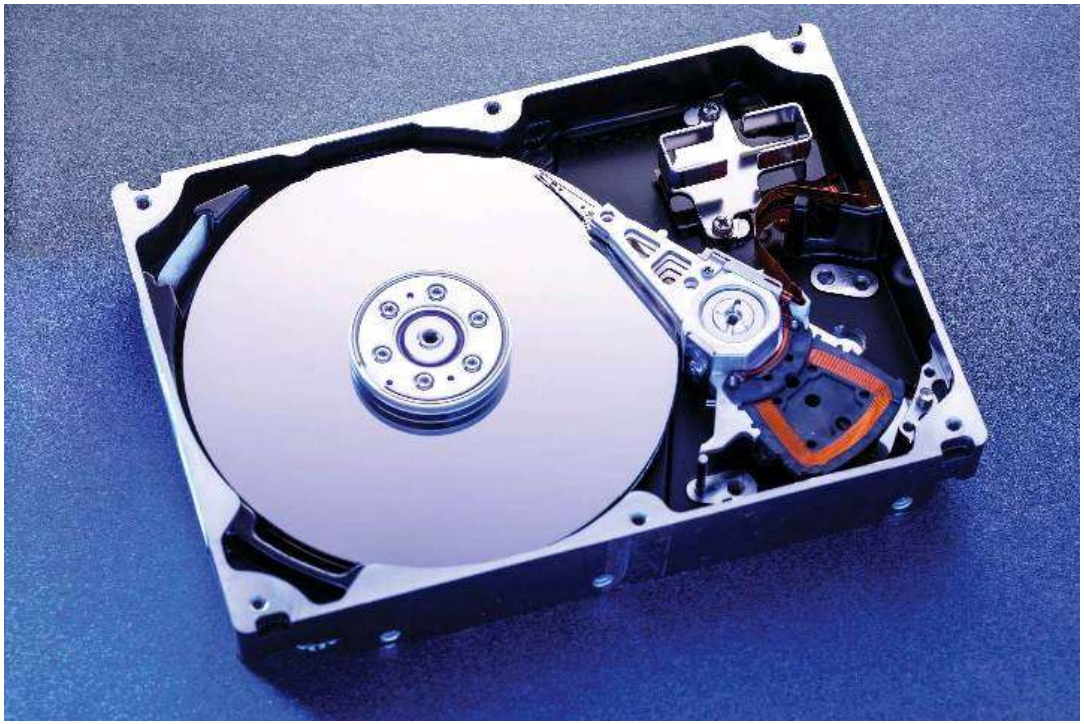
- **Head crash** results from disk head making contact with the disk surface
  - That's bad
- Disks can be removable

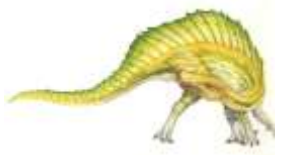




A 3.5" HDD with Cover Removed.

Magnetic disks





The First Commercial Disk Drive



1956
IBM RAMDAC computer included the IBM Model 350 disk storage system

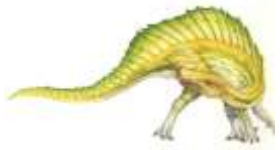
5 million (7 bit) characters
50 x 24” platters
Access time = < 1 second



Performance of Magnetic Disks

- Performance
  - Transfer Rate – theoretical – 6 Gb/sec
  - Effective Transfer Rate – real – 1Gb/sec
  - Seek time from 3ms to 12ms – 9ms common for desktop drives
  - Average seek time measured or calculated based on 1/3 of tracks
  - Latency based on spindle speed
    - 1 / (RPM / 60) = 60 / RPM
  - Average latency = ½ latency
- From Wikipedia

Spindle [rpm]	Average latency [ms]
4200	7.14
5400	5.56
7200	4.17
10000	3
15000	2





Mass-Storage Structure

Magnetic disks

Example

Calculate the average latency for a spindle speed 7200 RPM .

Solution

$$7200\,RPM = 60s$$

$$1RPM = \left(\frac{60}{7200}\right)/2$$

$$= 0.00833s/2$$

$$= 0.00417s = 4.17ms$$



Performance of Magnetic Disks (Cont.)

Magnetic disks

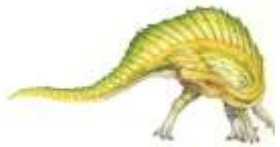
- **Access Latency** = **Average access time** = average seek time + average latency
  - For fastest disk 3ms + 2ms = 5ms
  - For slow disk 9ms + 5.56ms = 14.56ms
- Average I/O time = average access time + (amount to transfer / transfer rate) + controller overhead
- For example to transfer a 4KB block on a 7200 RPM disk with a 5ms average seek time, 1Gb/sec transfer rate with a .1ms controller overhead =
  - 5ms + 4.17ms + 0.1ms + transfer time
  - Transfer time = 4KB / 1Gb/s * 8Gb / GB * 1GB / 1024^2KB = 32 / (1024^2) = 0.031 ms
  - Average I/O time for 4KB block = 9.27ms + .031ms = 9.301ms





Secondary Storage Connection Methods

- Host-attached storage accessed through I/O ports talking to I/O busses
- Several bus technologies including **advanced technology attachment** (ATA), **serial ATA** (SATA), **eSATA**, **universal serial bus** (USB), **fibre channel** (FC), **serial attached SCSI** (SAS)
- Data transfers are carried out by special electronic processors: **controllers**
  - Host controller is in computer, device controller built into storage device
  - Talk to each other, usually via memory-mapped I/O ports
- Device controllers have built in caches
  - Data transfer from media into cache, then over bus to host (and vice versa)



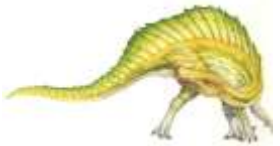


Solid-State Disks: SSD (NVM)

- Nonvolatile Memory (**NVM)** is electrical vs. mechanical (HDD)
- Commonly composed of a controller and flash NAND die semiconductor chips (A **die** is a small block of semiconducting material, on which a given functional circuit is fabricated
  - Other forms include DRAM with battery backup, non-NAND die like 3D XPoint
- Flash-memory-based NVM is frequently used in disk-drive like container -> **solid-state disk (SSD)**



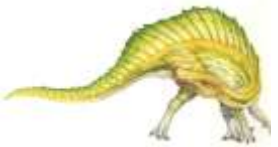
- Also can be in other formats like **USB drive**





SSD (Cont.)

- In all forms, acts and treated the same way
- More reliable than HDD (no moving parts), can be faster (no seek time or latency), consumes less power
- More expensive per MB, lower capacity, may have shorter lifespan (writes wear it out)
- Higher speed means new connection methods
  - Direct to PCIe bus (called NVMe)
- Can be used in variety of ways – replacement for disk, caching tier, etc

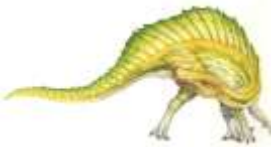




Magnetic Tape

- Was early secondary-storage medium
  - Evolved from open spools to cartridges
- Relatively permanent and holds large quantities of data
- Access time slow
- Random access ~1000 times slower than disk
- Mainly used for backup, storage of infrequently-used data, transfer medium between systems

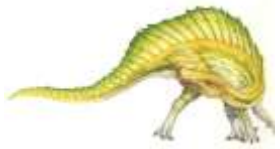






Magnetic Tape (Cont)

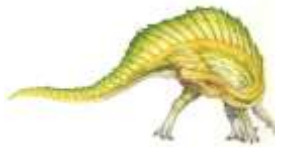
- Kept in spool and wound or rewound past read-write head
- Once data under head, transfer rates comparable to disk
  - 140MB/sec and greater
- GB to TB typical storage
- Common technologies are LTO-{5,6}, SDLT, 4, 8, and 19mm, and ¼ and ½" widths





Disk Scheduling

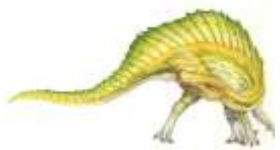
- The operating system is responsible for using hardware efficiently — for the disk drives, this means having a fast access time and disk bandwidth
- Minimize seek time
  - Seek time ≈ seek distance
- Disk **bandwidth** is the total number of bytes transferred, divided by the total time between the first request for service and the completion of the last transfer





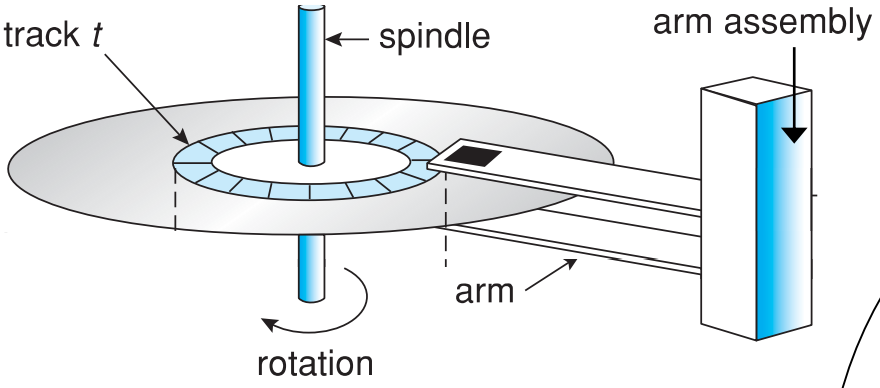
Disk Scheduling (Cont.)

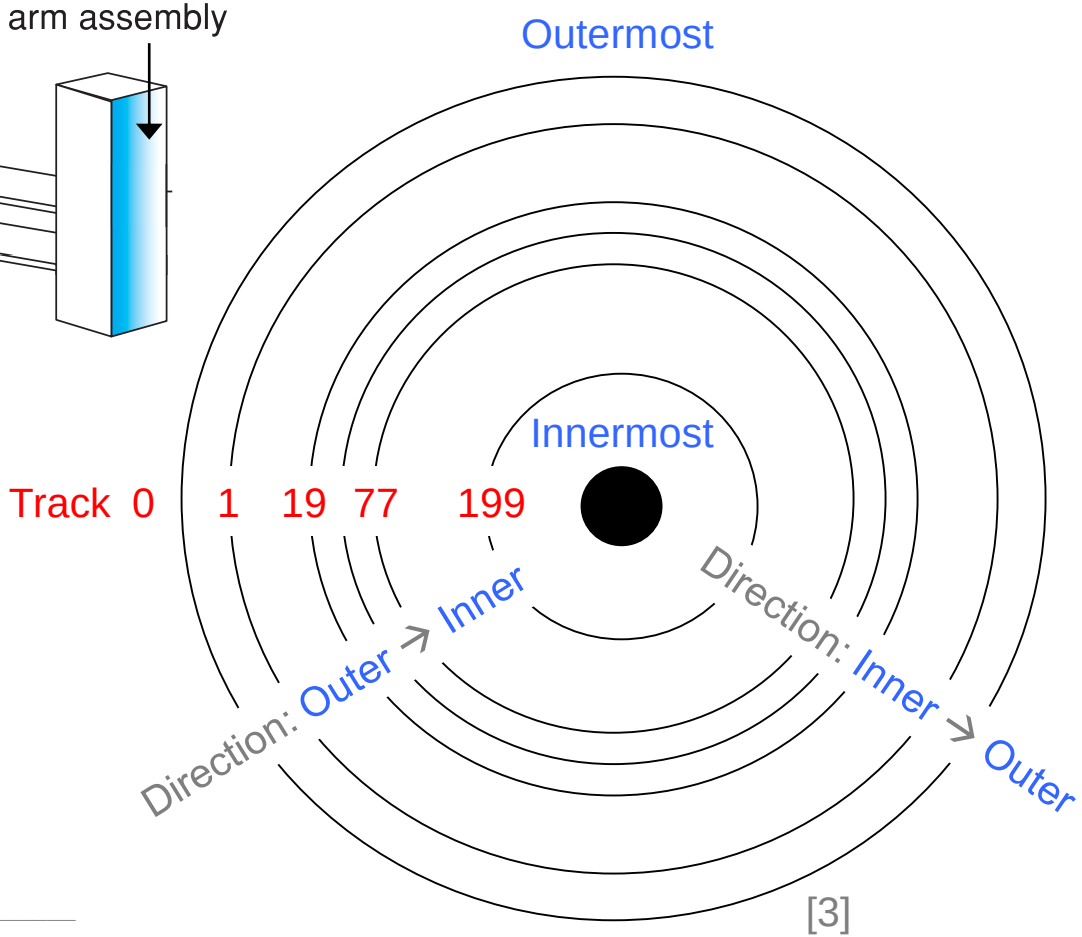
- There are many sources of disk I/O request
  - OS
  - System processes
  - Users processes
- I/O request includes input or output mode, disk address, memory address, number of sectors to transfer
- OS maintains queue of requests, per disk or device
- Idle disk can immediately work on I/O request, busy disk means work must queued.
  - **Optimization algorithms only make sense when a queue exists**





Disk Scheduling (Cont.)



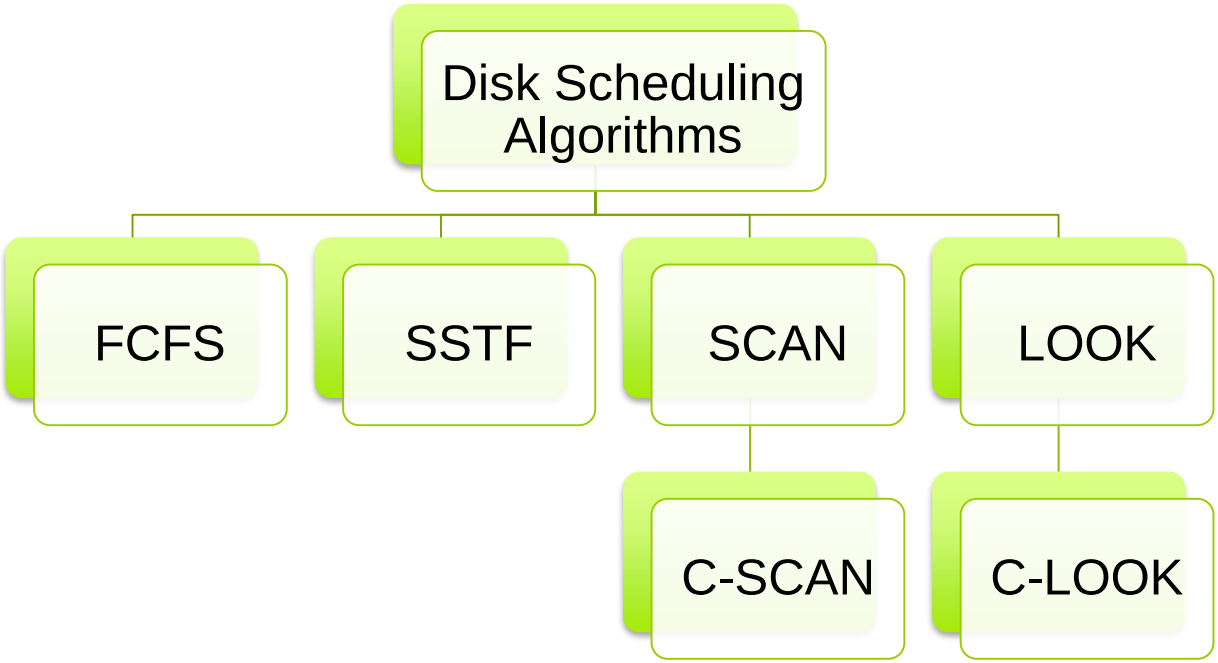


Resources:
[3] http://cnx.org/content/m27262/latest/



Disk Scheduling (Cont)

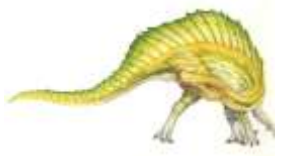
- Note that drive controllers have small buffers and can manage a queue of I/O requests (of varying “depth”)
- Several algorithms exist to schedule the servicing of disk I/O requests



First-Come, First-Serve (FCFS)
Shortest-Seek-Time-First (SSTF)
Circular SCAN (C-SCAN)
Circular LOOK (C-LOOK)



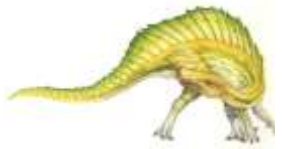
REFER TO PART B





Selecting a Disk-Scheduling Algorithm

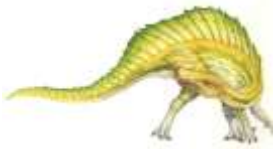
- SSTF is common and has a natural appeal
- SCAN and C-SCAN perform better for systems that place a heavy load on the disk
  - Less starvation
- Performance depends on the number and types of requests
- Requests for disk service can be influenced by the file-allocation method
  - And metadata layout





Disk-Scheduling Algorithm (Cont.)

- The disk-scheduling algorithm should be written as a separate module of the operating system, allowing it to be replaced with a different algorithm if necessary
- Either SSTF or LOOK is a reasonable choice for the default algorithm
- What about rotational latency?
  - Difficult for OS to calculate
- How does disk-based queuing effect OS queue ordering efforts?



Summary

Strategy	Advantages	Disadvantages
FCFS	• Easy to implement • Sufficient for light loads	• Doesn't provide best average service • Doesn't maximize throughput
SSTF	• Throughput better than FCFS • Tends to minimize arm movement • Tends to minimize response time	• May cause starvation of some requests • Localizes under heavy loads
SCAN/LOOK	• Eliminates starvation • Throughput similar to SSTF • Works well with light to moderate loads	• Needs directional bit • More complex algorithm to implement • Increased overhead
N-Step SCAN	• Easier to implement than SCAN	• The most recent requests wait longer than with SCAN
C-SCAN/C-LOOK	• Works well with moderate to heavy loads • No directional bit • Small variance in service time • C-LOOK doesn't travel to unused tracks	• May not be fair to recent requests for high-numbered tracks • More complex algorithm than N-Step SCAN, causing more overhead

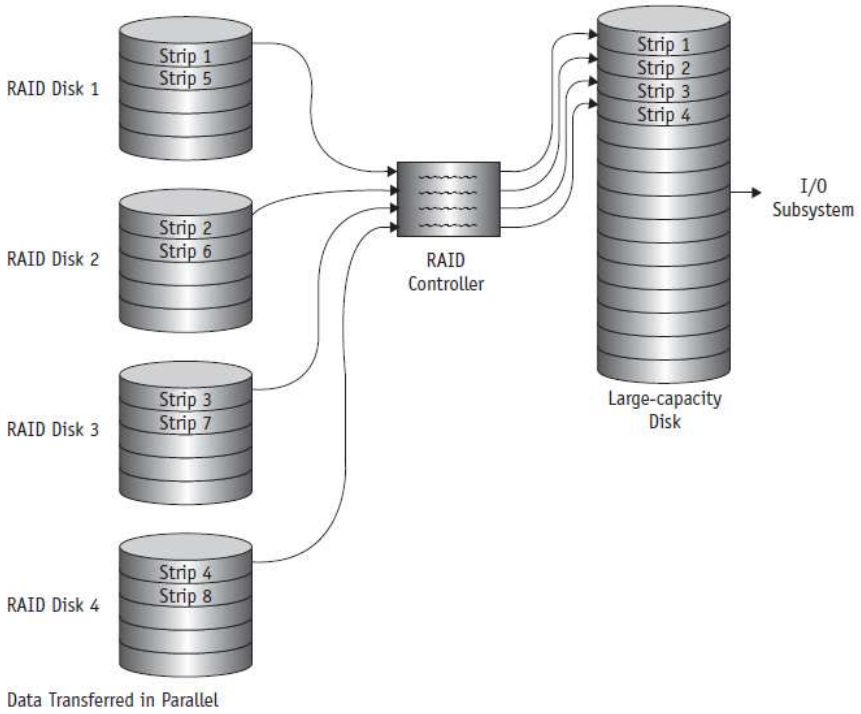
(table 7.9)

Comparison of DASD seek strategies discussed in this chapter.

RAID

- Stands for Redundant Arrays of Independent Disks
- Commonly used to address the performance and reliability issues.
- Frequently used on **servers**, but not generally used on PCs
- Normally consists of 2 or more disks working in parallel but appear as one drive to the user – **disk striping**.

- **Disk striping** can speed up data transfer because transfer can be done in parallel.



(figure 7.19)

Data being transferred in parallel from a Level o RAID configuration to a large-capacity disk. The software in the controller ensures that the strips are stored in correct order.

- RAID provides **reliability** through **redundancy**
  - *Mirroring* or *shadowing* keeps duplicate of each disk.
  - *Block interleaved parity* uses much less redundancy.

Data Striping:

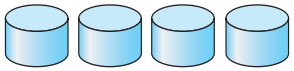
- A technique for spreading data over multiple disks
- Data striping speeds up operations that retrieve data from disk storage
- Data on disks are broken into units (blocks) and these are spread across the available disks
- Implementations allow selection of data units size, or *stripe width*

Data Mirrowing (Shadowing):

- A technique in which data are written to two duplicate disks simultaneously
- If one disk fails, the system can instantly switch to the other disk without loss of data or service
- Used commonly in on-line database systems where it is critical that data are accessible at all times



12.7.3 RAID Levels



(a) RAID 0: non-redundant striping.



(b) RAID 1: mirrored disks.



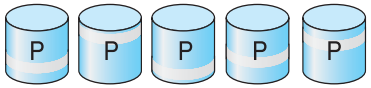
(c) RAID 2: memory-style error-correcting codes.



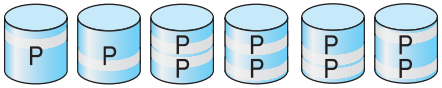
(d) RAID 3: bit-interleaved parity.



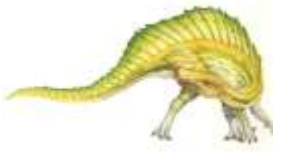
(e) RAID 4: block-interleaved parity.



(f) RAID 5: block-interleaved distributed parity.



(g) RAID 6: P + Q redundancy.

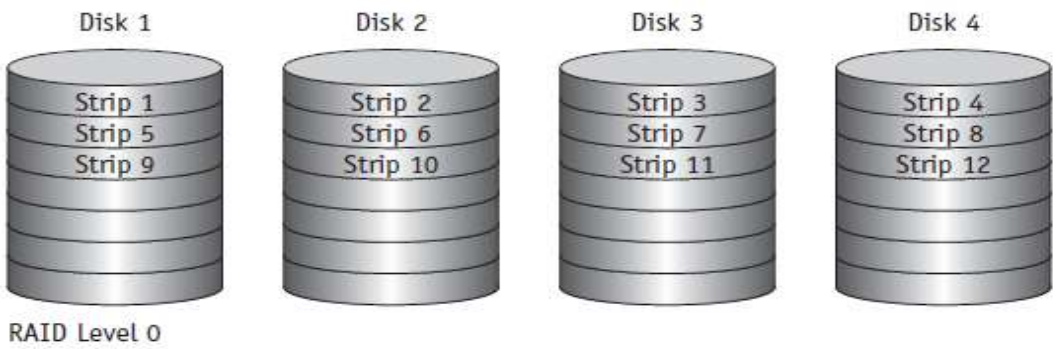


Level Zero

- Uses only data striping (not considered true RAID)
  - No parity and error corrections
  - No error correction/redundancy/recovery
- Benefits
  - Devices appear as one logical unit
  - Best for large data quantity non-critical data

(figure 7.20)

RAID Level 0 with four disks in the array. Strips 1, 2, 3, and 4 make up a stripe. Strips 5, 6, 7, and 8 make up another stripe, and so on.

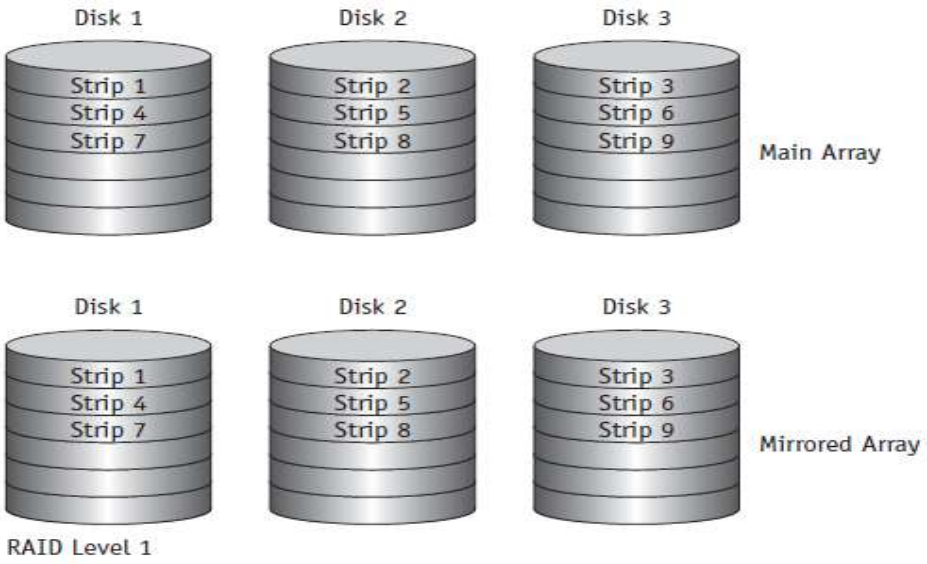


Level One

- Uses data striping and mirroring (considered true RAID)
  - Mirroring configuration (backup)
    - Duplicate set of all data (expensive)
  - Provides redundancy and improved reliability

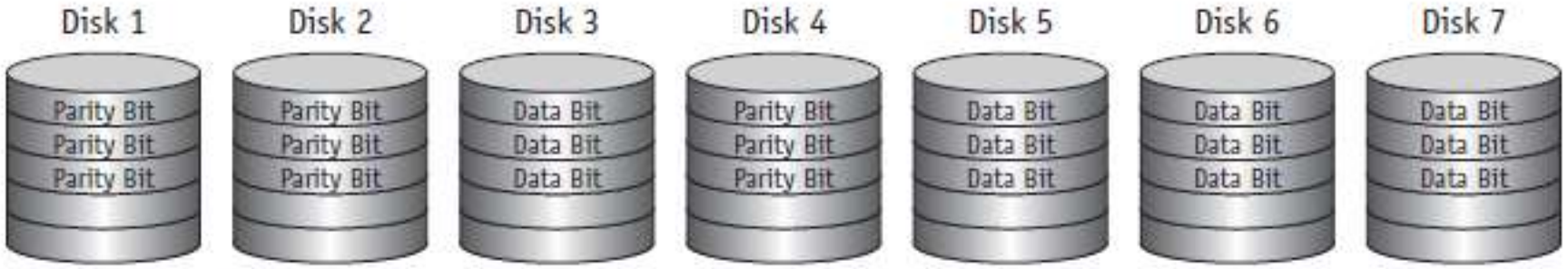
(figure 7.21)

RAID Level 1 with three disks in the main array and three corresponding disks in the backup array, the mirrored array.



Level Two

- Uses small stripes (bit level) (considered true RAID)
- **Hamming code**: error detection and correction
- Expensive and complex
  - Size of strip determines number of array disks



RAID Level 2 - Hamming Code

(figure 7.22)

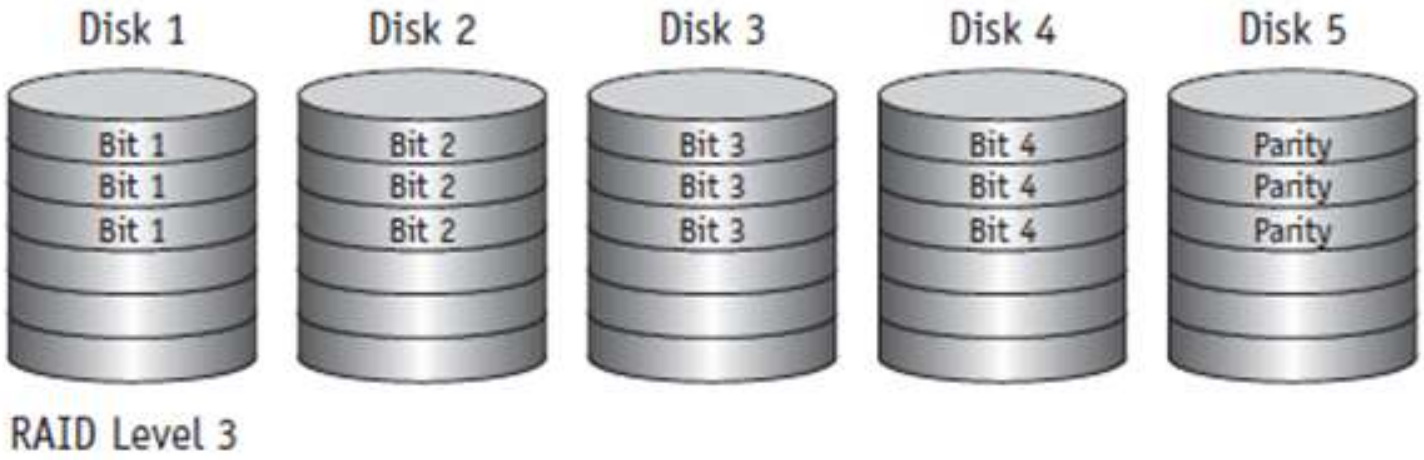
RAID Level 2. Seven disks are needed in the array to store a 4-bit data item, one for each bit and three for the parity bits. Each disk stores either a bit or a parity bit based on the Hamming code used for redundancy.

Level Three

- Modification of Level 2
  - Requires one disk for redundancy
    - Small strip (bit level) for data bits
    - Block strip for parity bits

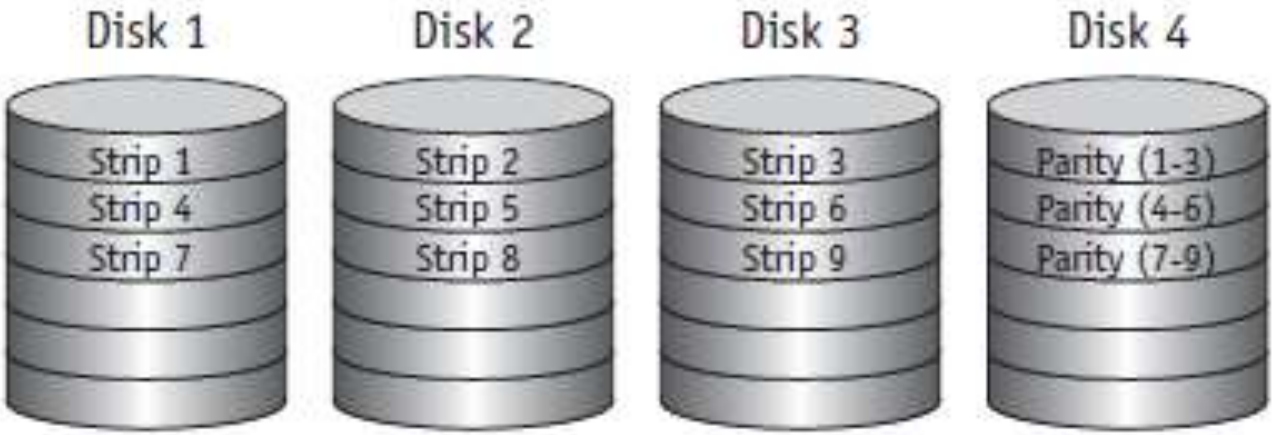
(figure 7.23)

RAID Level 3. A 4-bit data item is stored in the first four disks of the array. The fifth disk is used to store the parity for the stored data item.



Level Four

- Same strip scheme as Levels 0 and 1
  - Computes parity for each strip
  - Stores parities in corresponding strip
    - Has designated parity disk

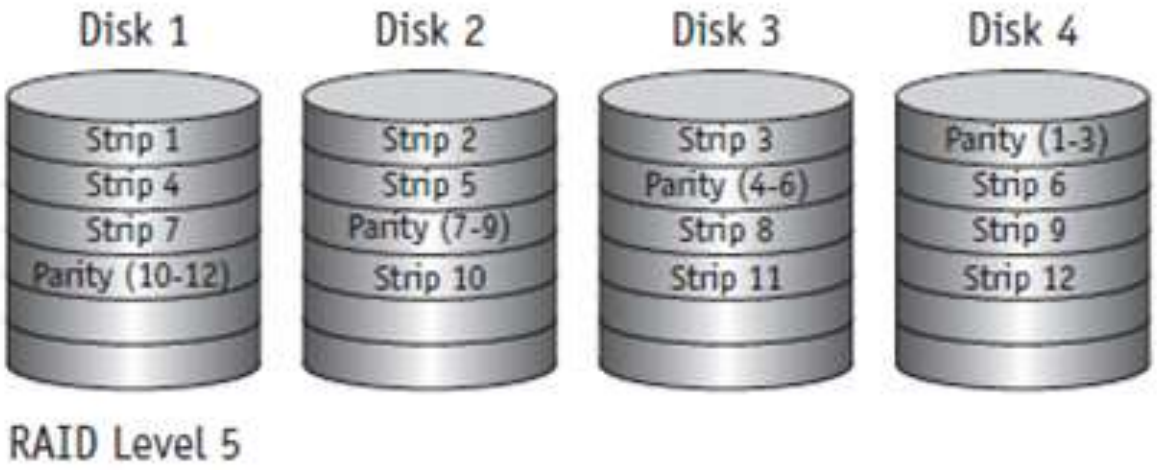


(figure 7.24)

RAID Level 4. The array contains four disks: the first three are used to store data strips, and the fourth is used to store the parity of those strips.

Level Five

- Modification of Level 4
- Distributes parity strips across disks
  - Avoids Level 4 bottleneck
- Disadvantage
  - Complicated to regenerate data from failed device

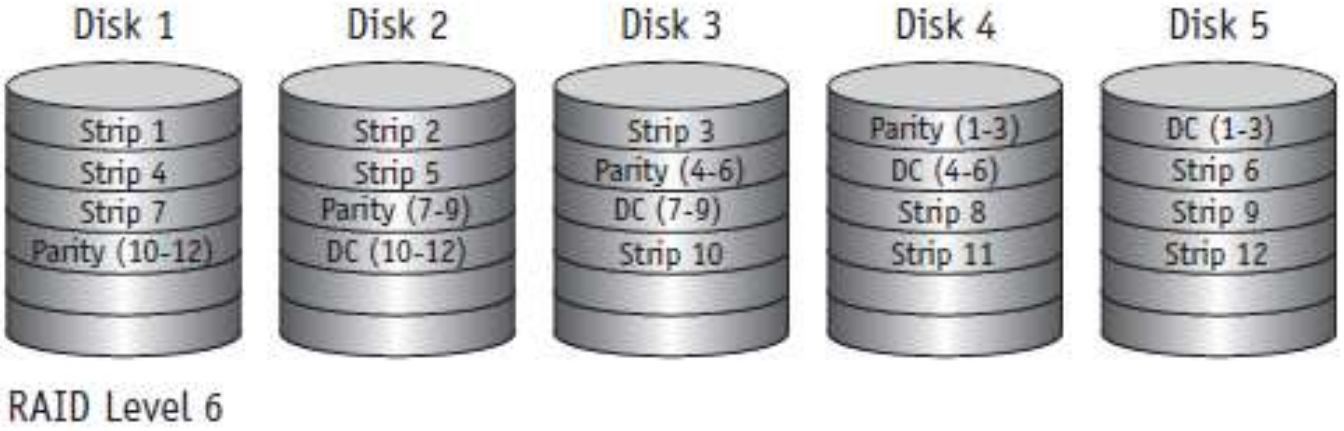


(figure 7.25)

RAID Level 5 with four disks. Notice how the parity strips are distributed among the disks.

Level Six

- Provides extra degree of error protection/correction
  - Two different parity calculations (double parity)
    - Same as level four/five and independent algorithm
  - Parities stored on separate disk across array
    - Stored in corresponding data strip
- Advantage: data restoration even if two disks fail



(figure 7.26)

RAID Level 6. Notice how parity strips and data check (DC) strips are distributed across the disks.

RAID

(table 7.7) *The seven standard levels of RAID provide various degrees of error correction. Cost, speed, and the system's applications are significant factors to consider when choosing a system.*

RAID Level	Error Correction Method	I/O Request Rate	Data Transfer Rate
0	None	Excellent	Excellent
1	Mirroring	Read: Good Write: Fair	Read: Fair Write: Fair
2	Hamming code	Poor	Excellent
3	Word parity	Poor	Excellent
4	Strip parity	Read: Excellent Write: Fair	Read: Fair Write: Poor
5	Distributed strip parity	Read: Excellent Write: Fair	Read: Fair Write: Poor
6	Distributed strip parity and independent data check	Read: Excellent Write: Poor	Read: Fair Write: Poor

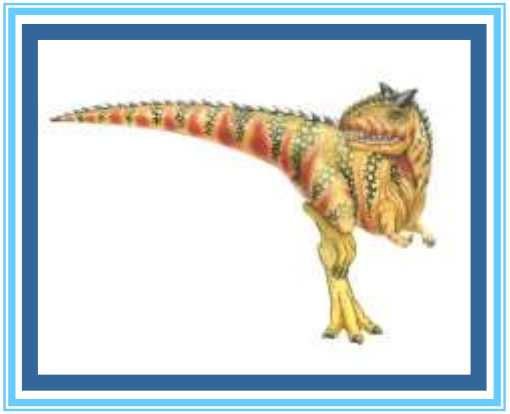
Summary

- Disk performance (disk access time) is based on seek time, latency and transfer time
- Disk scheduling algorithms
  - FCFS is a disk seek algorithm that is based on first come first served.
  - SSTF – serve request with track closest to one being served
  - SCAN and C-SCAN – serve request from outer to inner track and serve every request on way back to inner track. C-SCAN serves during inward sweep only
  - LOOK and C-LOOK – similar to SCAN and C-SCAN respectively except does not scan until the edge of cylinder

Summary

- RAID
  - Data striping is used to increase performance
  - Data mirroring and parity are techniques used for reliability since these techniques provide redundancy of data

End of Chapter 12



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